Pressure Analysis for Horizontal Wells

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Summary. This paper presents horizontal-well test design and interpretation methods. Analytical solutions are developed that can be handled easily by a desktop computer to carry out design as well as interpretation with semilog and log-log analysis. These analytical solutions point out the distinctive behavior of horizontal wells: (1) at early time, there is a circular radial flow in a vertical plane perpendicular to the well, and (2) at late time, there is a horizontal pseudoradial flow. Each type of flow is associated with a semilog straight line to which semilog analysis has to be adapted. The horizontal pseudoradial flow takes into account a pseudoskin depending on system geometry, which is *a priori* defined and estimated. Practical time criteria are proposed to determine the beginning and the end of each type of flow and to provide a guide to semilog analysis and well test design.

We study the behavior of uniform-flux or infinite-conductivity horizontal wells, with wellbore storage and skin. The homogeneous reservoir is infinite or limited by impermeable or constant-pressure boundaries. A method is also outlined to transform all our solutions for homogeneous reservoirs into corresponding solutions for double-porosity reservoirs.

Introduction

The first horizontal producing wells yielded extremely positive results,[1-4] which made it necessary to study *flow around a horizontal well* with a view to practical applications: test design and interpretation. The purpose of this study is to provide, with a horizontal well configuration, a way (1) to decide whether a test has to be made (will the test help to obtain the required information?); (2) to optimize test time, if necessary (the test must be long enough to be interpreted); and (3) to interpret the test performed with applicable methods.

The simplest and most significant analytical approach is first described in this paper. It relates to the transient behavior of a well with no wellbore storage, C, and no skin, s, with a uniform flow. (Flow per unit length is the same everywhere in the well.) Then the wellbore storage, the skin, and the effect of the boundaries of the reservoir are taken into account, leading to a practical approach applicable to most real cases. The transposition of these solutions to infinite-conductivity wells is described.

Horizontal Well in an Infinite Homogeneous Reservoir

The $2L$-long horizontal well is parallel to the impermeable boundaries of a reservoir of thickness h, conductivity k/μ, and capacity ϕc_t. The assumptions necessary for establishing and solving the diffusivity equation are the same as for the usual analytical solutions developed for well testing. Gravity forces are neglected and the reservoir is assumed to be isotropic with a single-phase liquid flow.

Well With No Skin and No Wellbore Storage (Infinite Reservoir). Let us assume that the well is horizontal with a radius r_w, has no skin, is located at a distance z_w from the lower boundary of an infinite reservoir (Fig. 1), and produces with a uniform flow and a constant bottomhole flow rate, qB (no wellbore storage).

Pressure-drop variation, Δp (equal to $p_i - p_{wf}$), with time at a point M of the well (determined by a value of y) was deduced with Green's method and source functions (Appendix A).[5,6] This solution can be written with the following dimensionless parameters:

$$p_D = a_1 \frac{kh\Delta p}{qB\mu}$$

and

$$t_D = a_2 \frac{kt}{\phi\mu c_t L^2},$$

where a_1 and a_2 are unit conversion factors (Table 1), and

$$e_{D3} = \frac{z_w}{h},$$

$$L_{D3} = \frac{L}{h},$$

$$y_D = \frac{y}{L},$$

and

$$x_{wD} = \frac{r_w}{L}.$$

The transient pressure behavior is a function of the following parameters:

$$p_D = p_D(t_D, y_D, e_{D3}, L_{D3}, x_{wD}).$$

The flow symmetry makes only one consideration possible: $e_{D3} > 0.5$ and $y_D > 0$.

Flow Types Around a Horizontal Well. The solution mentioned above allows plotting of the curves $p_D(t_D)$ for various values of parameters y_D, e_{D3}, and x_{wD} (discussed in the following three sections). All curves on semilog plots exhibit a common characteristic: they show two straight lines separated by a transition part. (Figs. 2 through 4 and Tables 2 through 4).

At early time, the pressure is given by

$$p_D = \frac{1}{2L_{D3}}\left[-\frac{1}{2}\mathrm{Ei}\left(-\frac{x_{wD}^2}{4t_D}\right)\right]$$

$$\cong \frac{1}{4L_{D3}}\left(\ln\frac{t_D}{x_{wD}^2} + 0.809\right) \text{ for } \frac{x_{wD}^2}{4t_D} < 10^{-2}. \quad \ldots\ldots\ldots (1)$$

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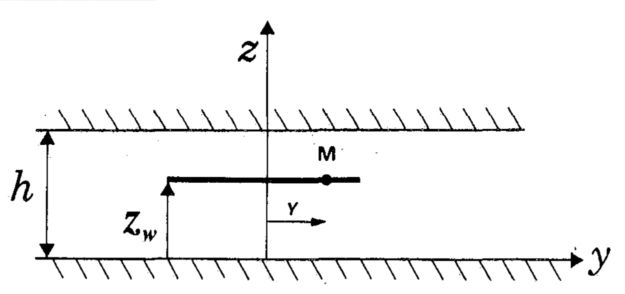

Fig. 1—Horizontal well in an infinite reservoir.

TABLE 1—UNIT CONVERSIONS

Symbol	Customary Units	SI Metric Units	Darcy Units
q	STB/D	stock-tank m^3/d	cm^3/s
μ	cp	cp	cp
k	md	md	darcies
h,L,r,y,z	ft	m	cm
p	psi	bars	atm
t	hours	hours	seconds
c_t	psi^{-1}	bar^{-1}	atm^{-1}
C	bbl/psi	m^3/bar	cm^3/atm
a_1	1/141.2	5.356×10^{-4}	2
a_2	2.64×10^{-4}	3.557×10^{-6}	1
a_3	0.89	0.159	1/2
b_1	162.6	2.149×10^3	
b_2	3.23	5.1	

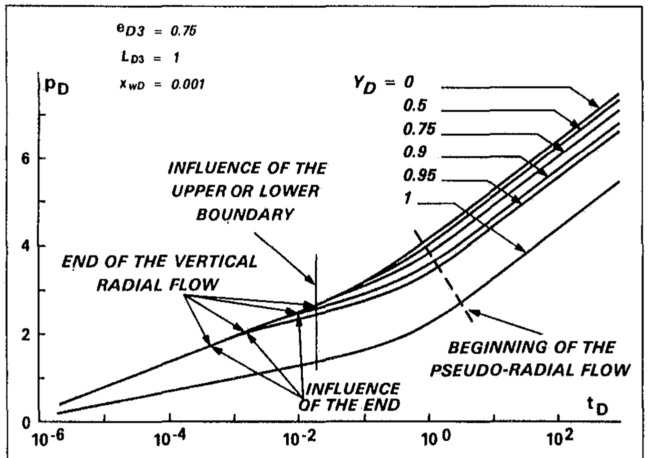

Fig. 2—Influence of y_D.

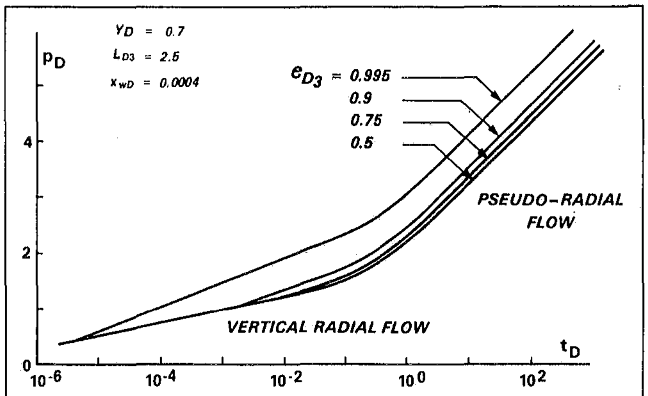

Fig. 3—Influence of e_{D3}.

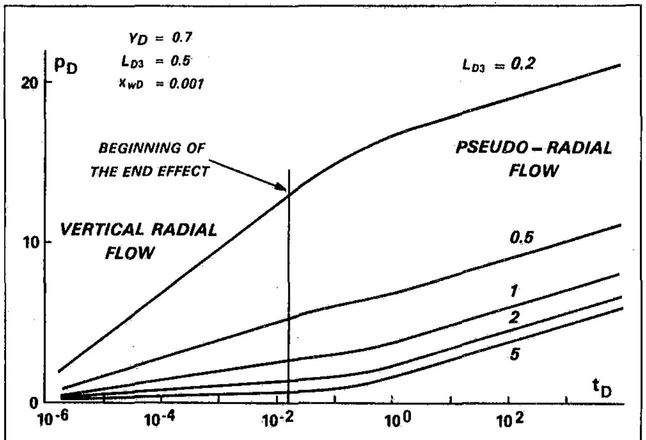

Fig. 4—Influence of L_{D3}.

TABLE 2—INFLUENCE OF y_D

Dimensionless Parameters	
e_{D3}	0.75
L_{D3}	1
x_{wD}	0.001
Example of Corresponding Real Values, m [ft]	
$2L$	200 [656]
h	100 [328]
z_w	75 [246]
r_w	0.1 [0.33]

A graph of p_D vs. log t_D yields a straight line of slope $1.15/2L_{D3}$. This solution corresponds to a *vertical radial flow* developing perpendicularly to the well axis (Fig. 5).

For $y_D=1$, the flow is never radial circular. At early time, the transient-pressure distribution is given in dimensionless form by the following equation:

$$p_D=\frac{1}{2L_{D3}}\left[-\frac{1}{2}\mathrm{Ei}\left(-\frac{x_{wD}^2}{4t_D}\right)\right]$$

$$\cong\frac{1}{8L_{D3}}\left(\ln\frac{t_D}{x_{wD}^2}+0.809\right)\text{ for }\frac{x_{wD}^2}{4t_D}<10^{-2}.\ \ldots\ldots\ldots(2)$$

Therefore, when $y_D=1$, the slope of the first semilog straight line is one-half the one observed for the other values of y_D (Eq. 1), everything being equal in other parts.

The vertical radial flow vanishes when the effect of reservoir boundaries or of well ends is perceptible.

1. The nearest vertical boundary affects well pressure, when reached by the radius of investigation of the vertical radial flow. The vertical radial flow stops when $t_D\cong0.32\ [(1-e_{D3})/L_{D3}]^2$ (Appendix A).

2. The end effect is noticeable when $t_D\cong(1-y_D)^2/6\ (y_D\neq1)$.

These time criteria give only orders of magnitude. These are quite helpful, however, because they refer to values really observed on the pressure curves. Of course, the criterion from the investigation radius (Appendix A) is valid only if the end effect has not caused distortions in the lines of flow.

When t_D increases, the real curve deviates from the first straight line. Let us consider the curves for which $L>h$—i.e., those of Figs. 2 and 3. It appears from these curves that pressure p_D can be different during the transition period following a vertical radial flow.

1. The effect of the nearest vertical boundary appears first and the real curve passes above the first straight line (slope increase).

2. The end effect appears first and the real curve is under the first straight line (slope decrease).

When t_D increases, equipotential surfaces become more complex in shape because of the effect of ends and boundaries. The more

TABLE 3—INFLUENCE OF e_{D3}	
Dimensionless Parameters	
y_D	0.7
L_{D3}	2.5
x_{wD}	0.0004
Example of Corresponding Real Values, m [ft]	
$2L$	500 [1,640]
h	100 [328]
r_w	0.1 [0.33]

distant from the well, the more similar to cylinders these surfaces are. After a long enough time, flow is almost incompressible between the horizontal well and the first cylinder-like equipotential surface. From that time on, well pressure variation depends only on the transient fluid flow through the reservoir area beyond the cylindrical equipotential surface; flow is radially circular in a horizontal plane (Fig. 5). This flow is called pseudoradial flow. The corresponding pressure behavior may be given by the following equation:

$$p_D(t_D)=0.5(\ln\ t_D+\alpha), \quad (3)$$

where α involves a geometrical skin, as discussed in Pseudoradial Flow. The pseudoradial flow begins at a time t_D within the following time range: $0.8<t_D<3$ (Figs. 2 through 4).

Well Measurement Point (y_D Effect). As Fig. 2 and Table 2 show, when y_D increases, (1) the end effect appears earlier and pressure drop is lower, (2) the time at which the nearest boundary effect is noticeable is unchanged, and (3) the pseudoradial flow begins slightly later (from $t_D=0.8$ to $t_D=3$).

The infinite-conductivity approximation relates more closely to the real case than the uniform-flow approximation used here. With the infinite-conductivity approximation, the pressure is the same everywhere in the horizontal well.

As indicated in Approximate Solution for a Horizontal Well With Infinite Conductivity, the "uniform flow" solution with $y_D=0.7$ enables, in most cases, the proper simulation of the infinite-conductivity well. For this reason, in the following paragraphs we will use only $y_D=0.7$.

Off-Center (e_{D3}) Effect. Curves in Fig. 3 (Table 3) show that the vertical radial flow duration is reduced when the well is out of the reservoir center. When e_{D3} increases, the nearest boundary effect is felt earlier. In this figure, the beginning of the pseudoradial flow is not very sensitive to e_{D3} and is in the range of $t_D=1.9$ in the example of Fig. 3.

The curve $e_{D3}=0.995$ is identical with others for $t_D<2\times10^{-6}$ because there is still no perceptible effect of the boundary on well pressure, though it is very close (0.5 m [1.64 ft]).

Then curve $e_{D3}=0.995$ becomes, on a semilog plot, a straight line with a slope of $1.15/L_{D3}$ instead of $1.15/(2L_{D3})$, as in the case of a vertical radial flow. The doubling of slope value can be accounted for by analogy with a vertical well close to a linear fluid barrier.

Applying the method of images,

$$p_D=-\frac{1}{4L_{D3}}\left\{\mathrm{Ei}\left(-\frac{x_{wD}^2}{4t_D}\right)+\mathrm{Ei}\left[-\frac{(1-e_{D3})^2}{L_{D3}^2t_D}\right]\right\},$$

which in our example can soon be approached as

$$p_D=\frac{1}{2L_{D3}}\left[\ln\frac{t_D}{x_{wD}^2}+0.809+\ln\frac{L_{D3}x_{wD}}{2(1-e_{D3})}\right]. \quad (4)$$

This represents the equation of a semilog straight line with a slope $1.15/L_{D3}$. The pseudoradial flow then begins for $t_D>1.9$.

Influence of L_{D3}. The length of the well relative to the depth of the reservoir ($2L_{D3}$) has an influence during the vertical radial flow: the slope of the corresponding semilog straight line is $1.15/2L_{D3}$ (Fig. 4). This parameter does not have any influence on the pseudoradial flow: the slope of the corresponding semilog straight line is 1.15 [see Well Measurement Point (y_D Effect)]. The time t_D corresponding to the beginning of the pseudoradial flow is influenced very little by L_{D3} (Fig. 4).

Well With Skin and No Wellbore Storage (Infinite Reservoir). By analogy with vertical wells, the skin of a horizontal well is defined, considering that skin pressure drop is constant for a constant bottomhole flow (no wellbore storage).

$$\Delta p_s=\frac{qB\mu}{2\pi k2L}s. \quad (5)$$

With these assumptions, for a well with no wellbore storage, the dimensionless pressure, p_D, of a horizontal well with a skin can be obtained from horizontal well dimensionless pressure, p_D, without skin ($s=0$) by

$$p_D=p_D(s=0)+\frac{1}{2L_{D3}}s. \quad (6)$$

Another way to take the skin into account is the method of the effective radius,[6,7] which involves replacing the actual well with a radius r_w and a skin s by an equivalent well with a radius $r_w'=r_we^{-s}$ and no skin. This is the method we use because it is the only one that is applicable to all cases, especially for a well with wellbore storage and negative skin.

Well With Wellbore Storage and No Skin (Infinite Reservoir). Wellhead flow rate variations generally are the only known. Wellbore storage causes the bottomhole flow rate to change more slowly than at the wellhead. In our analytical model, wellbore storage effect is taken into account with the Cinco-Ley and Samaniego[8] model. Adjusting this simulator to the solution of Appendix A requires determining a dimensionless wellbore storage, C_{De}, for a horizontal well:

$$C_{De}=a_3\frac{C}{L^2h\phi c_t},$$

where a_3 is a unit conversion factor (Table 1).

The semilog plot of Fig. 6 ($s=0$) shows that wellbore storage affects mainly the vertical radial flow. Therefore, wellbore storage has an effect essentially when the horizontal well behaves like a

TABLE 4—INFLUENCE OF L_{D3}

Dimensionless Parameters			Example of Corresponding Real Values			
			$2L$		r_w	
y_D	e_{D3}	x_{wD}	m	ft	m	ft
0.7	0.5	0.001	200	656	0.1	0.33

L_{D3}	0.2	0.5	1	2	5
h					
m	500	200	100	50	20
ft	1,640	656	328	164	65.6

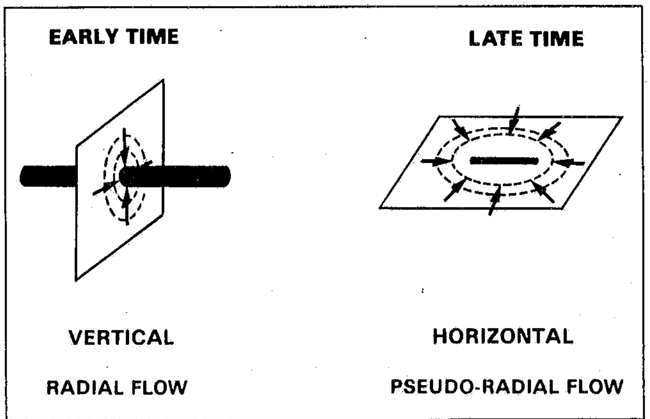

Fig. 5—Flow types around a horizontal well.

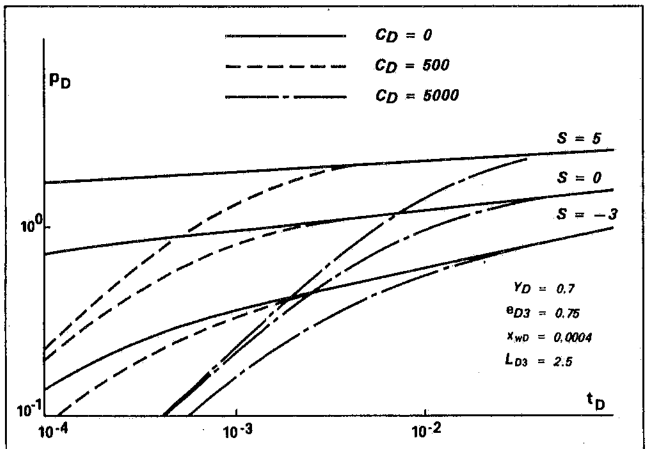

Fig. 6—Log-log plot—early time—influence of C_D.

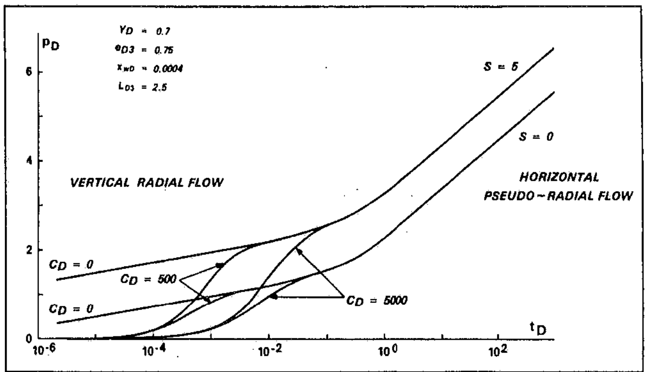

Fig. 7—Influence of C_D.

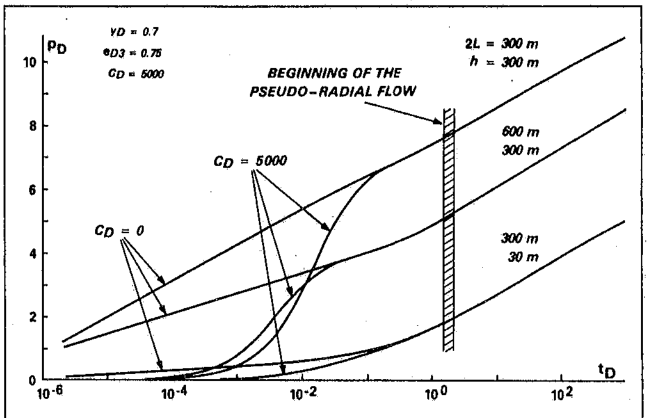

Fig. 8—Beginning of the pseudoradial flow.

TABLE 5—TIME CRITERIA (t_D)

Beginning	End
Vertical Radial Flow	
End of wellbore storage effect	$\min\left[\dfrac{(1-y_D)^2}{6};\ 0.32\left(\dfrac{1-e_{D3}}{L_{D3}}\right)^2\right]$
Pseudoradial Flow	
1.5 to 2	$\min\left[0.4\left(\dfrac{1-e_{D1}}{L_{D1}}\right)^2;\ 0.4\left(\dfrac{1-e_{D2}}{L_{D2}}\right)^2\right]$

vertical well in a $2L$-thick reservoir—i.e., like a vertical well with a wellbore storage:

$$C_D=a_3\frac{C}{r_w^2 2L\phi c_t}.$$

For an easier comparison with vertical well curves, the C_D values associated with the C_{De} ones used in the model are mentioned on every curve of Figs. 6 through 8.

Well With Wellbore Storage and Skin (Infinite Reservoir). This general solution is obtained from the analytical solution described in Appendix A with the method of effective radius and the wellbore storage simulator. This solution was compared during the vertical radial flow with the one relating to a vertical well with wellbore storage and skin as presented in Ref. 7 type curves. As expected, the difference between these solutions is the same as the one between the "line source" solution (our solution) and the "cylindrical well" solution (Ref. 7 solution). Our solution—shown in Fig. 6 for small time values—fits the Ref. 7 solution for positive, null, or slightly negative skins. But a highly negative skin ($s=-3$) introduces a very great effective radius in both solutions, which therefore differs for small time values ($t_D<1.2\times10^{-3}$). This difference, maximum for a well with no wellbore storage, C_D, decreases when C_D increases (real cases). It is negligible for all realistic time values when C_D=5,000.

First Semilog Straight Line. The analytical solution is plotted vs. time on a log-log plot in Fig. 6 with values of s=5 and s=0. (For $s=-3$, the curve is too flat.) Fig. 7 shows that the first semilog straight line almost always disappears because of the wellbore storage effect.

The semilog analysis of this first straight line would be possible only for highly negative skins and small wellbore storage values (short duration wellbore storage effect), provided that the criteria mentioned in Flow Types Around a Horizontal Well allow this. In most real cases, the first semilog straight line is unlikely to appear.

Second Semilog Straight Line. Let us assume that the pseudoradial flow begins when the slope of the curve differs <10% from the slope of the corresponding semilog straight line. Then Fig. 7 shows that the pseudoradial flow is never disturbed by wellbore storage effect. In Fig. 7, several values are assigned to L_{D3}—i.e., 5, 1, and 0.5—because it is the only parameter being acted upon at the beginning of the pseudoradial flow start.

As a conclusion, it can be stated that for all realistic configurations, pseudoradial flow can be observed for t_D ranging between 1.5 and 2.

As an illustration,

$2L$=400 m [1,312 ft],

$\phi c_t=10^{-7}$ kPa^{-1} [1.45×10^{-8} psi^{-1}],

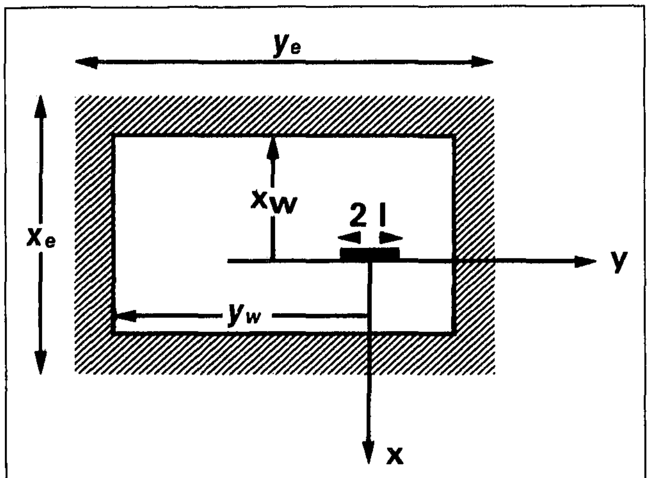

Fig. 9—Horizontal well in a limited reservoir.

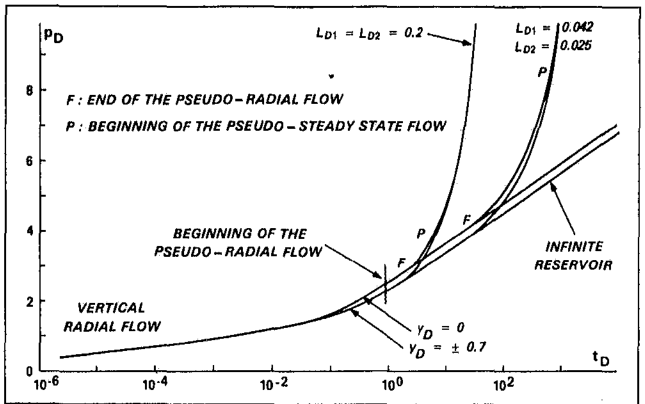

Fig. 10—Horizontal well in a reservoir limited by impermeable boundaries.

and

$k/\mu = 10$ md/(Pa·s) [10×10^3 md/cp].

Pseudoradial flow starts at a time t of between 7.1 and 9.5 days. For a semilog analysis, one should wait until a time at least threefold longer. For small conductivity values and long wells, testing time is extremely long. This criterion is essential to determine the minimum time required for a test to be interpretable.

Horizontal Well Test Interpretation. ***General.*** Let us consider a horizontal well producing with a constant wellhead flow rate and located on a reservoir behaving like an infinite reservoir. Flowing pressures can be interpreted with the same methods as for conventional analysis: log-log and semilog analysis.

Two categories of flows corresponding to semilog straight lines were seen. The main problem concerns the positioning of these straight lines when they exist. For this purpose, we set out very simple time criteria (see Flow Types Around a Horizontal Well, Second Semilog Straight Line, and Table 5). The period during which these wellbore storage effects were perceptible, however, remains difficult to appreciate. The best procedure consists in (1) generating type curves with the analytical solution proposed in Appendix A and (2) varying unknown parameters to generate a type curve that fits the experimental result. Such an analytical simulation will help to detect eventual semilog straight lines and will lead to a first-parameter estimation and to a better understanding of phenomena. In each case, semilog analysis will be a basis for interpretation and the most readily applicable method.

Semilog Analysis. Vertical Radial Flow. Assuming that the wellbore storage effect is not long enough to mask the first semilog straight line, the equation of the corresponding straight line can be written by combining Eqs. 1 and 5 or 1 and 6 as follows:

$$p_i - p_{wf}(t) = \frac{qB\mu}{4\pi k 2L}\left(\ln\frac{kt}{\phi\mu c_t r_w^2} + 0.809 + 2s\right). \quad (7)$$

Therefore, the conventional semilog analysis can be applied here by replacing thickness h with well length $2L$. The value of $k2L$ can be calculated from the slope m of the semilog straight line. The average permeability observed during a circular radial flow in a vertical plane normal to the well can be deduced from it. The equations are as follows:

$$k = -\frac{b_1 qB\mu}{m2L}$$

and

$$s = 1.15\left(\frac{p_{1h} - p_i}{m} - \log\frac{k}{\phi\mu c_t r_w^2} + b_2\right),$$

where b_1 and b_2 are unit conversion factors (Table 1).

Pseudoradial Flow. If the test is long enough and if boundary effect is not felt too early, a pseudoradial flow can be observed. The equation of the corresponding semilog straight line can be written by combining Eqs. 5 and A-4 as follows:

$$p_i - p_{wf}(t) = \frac{qB\mu}{4\pi kh}\left(\ln\frac{kt}{\phi\mu c_t r_w^2} + 0.809 + 2s'\right) \quad (8)$$

with

$$s' = (h/2L)s + s_g, \quad (9)$$

where s_g is the pseudoskin or geometrical skin corresponding to the pressure-drop reduction occurring when a vertical well is substituted by a horizontal well. The geometrical skin is negative and in practical cases can be determined by application of Eq. A-6 (Appendix A):

$$s_g = \ln\frac{2x_{wD}}{(2\pi x_{wD} L_{D3})^{1/(2L_{D3})}}.$$

In the case illustrated in Figs. 6 and 7, the calculated value for s_g is -6.1.

Eq. A-6 appears to be less accurate for a significant off-centering, as indicated in Appendix A. The conventional equation can be directly applied to obtain (1) the value for kh, where k is the average remote permeability governing the horizontal circular radial flow, and (2) the value for a global skin s', from which s later can be determined by application of Eq. 8. The geometrical skin, s_g, has been previously calculated with Eq. A-6.

Horizontal Well in a Homogeneous Reservoir With Impermeable Boundaries

We will now describe the phenomena related to boundary effects—i.e., those occurring at the end of the transient flow. To simplify the approach, let us neglect skin and wellbore storage, whose effects are well-known.

The position of the horizontal well in the reservoir, which is assumed to be limited by rectangular impermeable boundaries, is schematized in Fig. 9. This configuration is determined by the following four additional parameters: $L_{D1} = L/x_e$, $L_{D2} = L/y_e$, $e_{D1} = x_w/x_e$, and $e_{D2} = y_w/y_e$.

The pressure behavior for a uniform flow is given by (Appendix B) $p_D = p_D(t_D, y_D, L_{D1}, L_{D2}, L_{D3}, e_{D1}, e_{D2}, e_{D3}, x_{wD})$.

By reason of symmetry and taking some precautions (Appendix B), we can limit the study to dimensionless off-center values $e_D > 0.5$.

The solution for infinite-conductivity wells is still identical to the uniform flow solution with $y_D = \pm 0.7$ for $e_{D2} = 0.5$ (see Appendix B).

TABLE 6—HORIZONTAL WELL IN RESERVOIR LIMITED BY IMPERMEABLE BOUNDARIES

Dimensionless Parameters

	y_D	e_{D3}	L_{D3}	c_D	s	e_{D1}	e_{D2}	L_{D1}	L_{D2}	x_{wD}
Common parameters	0 ±0.7	0.75	2.5	0	0					
Configuration 1						0.5	0.5	0.2	0.2	0.0004
Configuration 2						0.75	0.5	0.042	0.025	−0.0004
Configuration 3						Infinite Reservoir				0.0004

Example of Corresponding Values

	2L (m)	2L (ft)	h (m)	h (ft)	z_w (m)	z_w (ft)	r_w (m)	r_w (ft)	x_e (m)	x_e (ft)	y_e (m)	y_e (ft)	x_w (m)	x_w (ft)	y_w (m)	y_w (ft)
Common parameters	500	1,640	100	328	75	246	0.1	0.33								
Configuration 1									1250	4,101	1250	4,101	625	2,051	625	2,051
Configuration 2									6000	19,685	10 000	32,808	4500	14,764	5000	16,404
Configuration 3									Infinite reservoir							

The solution described in Appendix B was tested with $y_e=2L$, $h=x_e$, and $e_{D1}=e_{D2}=e_{D3}=0.5$. This solution is obtained for a vertical well in the center of a $2L$-thick reservoir limited by impermeable boundaries forming a square (with side $x_e=h$).[7] At early time, the well behaves as in the case of an infinite reservoir because boundary effects are no longer felt (curves are identical in Fig. 10).

After a long enough time, p_D varies proportionally to t_D and the corresponding straight line has a slope

$$\frac{dp_D}{dt_D}=2L_{D1}L_{D2}$$

or

$$\frac{dp}{dt}=-\frac{qB}{x_e y_e h\phi c_t}.$$

This slope characterizes the pseudosteady-state flow. Curves for $y_D=\pm 0.7$ are identical because $e_{D2}=0.5$.

A significant phenomenon appears on the curves of Fig. 10 (Table 6): the semilog straight line corresponding to the pseudoradial flow does not exist in the case of Configuration 1 (Table 6) because the transient flow vanishes too early under the effect of boundaries. Because the vertical radial flow is frequently masked by the wellbore storage effect, tests performed under the conditions of Configuration 1 (Table 6) may be impossible to interpret because no semilog analysis can be made.

Before deciding on a horizontal well test, it is necessary to determine preliminarily the time at which the pseudoradial flow is expected to end. For this purpose, the best method is to estimate the pressure evolution by generating the corresponding type curve with the analytical solution described in Appendix B, as in the case of Fig. 10.

We also looked for a criterion that would enable us to estimate roughly and rapidly the end of the pseudoradial flow. For this purpose, we plotted several curves using the solution (Eq. A-6) with various values for L_{D1}, L_{D2}, e_{D1}, e_{D2}, and y_D.

At time t corresponding to the end of the pseudoradial flow, we have associated the dimensionless time t_{D0} not to L^2, but to the square of the distance from the horizontal well center to the nearest boundary. This boundary can be either parallel or perpendicular to the well.

Accordingly,

$$t_{D0}=\frac{L_{D1}^2}{1-e_{D1}}t_D$$

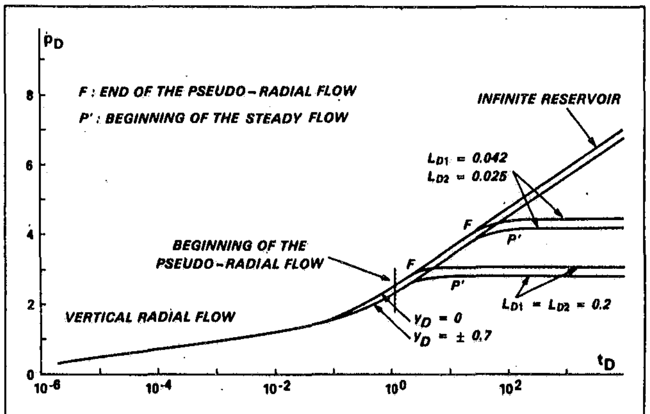

Fig. 11—Horizontal well in a reservoir limited by constant-pressure boundaries.

or

$$t_{D0}=\frac{L_{D2}^2}{1-e_{D2}}t_D$$

for $e_{D1}>0.5$, $e_{D2}>0.5$.

We obtained, for all studied configurations, t_{D0} values ranging from 0.3 to 0.5. We will now use 0.4 for t_{D0}. For configurations 1 and 2 of Table 6, the values obtained for the end of the pseudoradial flow are $t_{D1}=2.5$ and $t_{D2}=14.3$, respectively. These values are in agreement with those of Fig. 10. For example, with

$2L=400$ m [1,312 ft],

$\phi c_t=10^{-7}$ kPa^{-1} [0.7×10^{-7} psi^{-1}],

and

$k/\mu=10$ md/(mPa·s) [10×10^3 md/cp],

the values of t_{D1} and t_{D2} correspond to a real time of 12 and 68 days, respectively.

Horizontal Well in a Homogeneous Reservoir With Constant-Pressure Boundaries

The assumptions and configurations dealt with below are the same as those described above (Fig. 9). The rectangular boundary is kept at a constant pressure. The analytical solution described in Appendix C allows us to generate the curves illustrated in Fig. 11 with the parameter values of Table 6. The beginning of the steady flow appears clearly on these curves.

The conclusions and criteria set out in Horizontal Well in a Homogeneous Reservoir With Impermeable Boundaries relating to the pseudoradial flow end remain unchanged. They are directly applicable in the case of constant-pressure boundaries.

Approximate Solution for a Horizontal Well With Infinite Conductivity

In the model we have used, the flow penetrating in the well is assumed to be uniform; this means that the flow penetrating into the well by surface unit is the same everywhere in the well. Actually, we may state only that the well permeability is greater than that of the reservoir (infinite-conductivity approximation). With this assumption, we can consider that pressure is the same everywhere in the well. To compare these two approaches, we have determined the equivalent point in the well with the uniform-flow approximation, where the pressure is the same as in an infinite-conductivity well.

In the model used for determining the equivalent point y_D, the horizontal well is simulated by point sources. Assigning a flow value for each point source allows us to simulate either an infinite-conductivity well or a uniform-flow well. It appears from all the cases dealt with that the determined value y_D ranges between 0.67 and 0.72 for any possible length of horizontal wells. Additionally, uniform flow curves for both values of y_D are very close. Therefore, the uniform flow solution with $y_D=0.7$ is a good approximation to simulate an infinite-conductivity well behavior. Note that this result is valid during transient flow (i.e., until the end of the pseudoradial flow) and can be theoretically extrapolated to limited reservoirs only with $e_{D2}=0.5$ (Appendix B).

Horizontal Well in a Reservoir With Double Porosity

A double-porosity reservoir can be treated as a homogeneous reservoir whose conductivity, k/μ, is kept constant and equal to that of fractures and whose capacity, ϕc_t, varies gradually from the fracture's capacity to that of the fracture-plus-matrix system.

The ϕc_t variation law is known in the Laplace domain.[6] From any solution applicable to a homogeneous reservoir, a solution can be determined for a double-porosity reservoir by a simple transformation through the Laplace domain.

Conclusions

We studied the behavior of a horizontal well with uniform flow or infinite conductivity with wellbore storage and skin in an infinite or limited reservoir (impermeable boundaries or constant-pressure boundaries). Analytical solutions readily usable by a minicomputer were developed. The resultant analytical model is a highly valuable tool for horizontal-well test design and interpretation.

Two types of flows occurring successively in horizontal wells showed up: vertical radial flow and horizontal pseudoradial flow. These two flows correspond in favorable cases to interpretable semilog straight lines to which conventional semilog analysis can be applied (with a geometrical skin).

Practical time criteria (Table 5) were established to determine the end and start of these flows for which a semilog analysis is possible.

In practice, the initial vertical radial flow will often be screened by wellbore storage effect. Therefore, it will be necessary before deciding to perform a horizontal-well test to make sure a pseudoradial flow exists—i.e., the reservoir boundaries will not affect well pressure too early and test duration will not be prohibitive. If these conditions are not verified, the test may be impossible to interpret. A check can be made with the criteria shown in Table 5 or with the analytical solutions.

Nomenclature

a_1,a_2,a_3
b_1,b_2 = unit conversion factors
B = reservoir volume factor
c_t = total isothermal compressibility, kPa^{-1} [psi^{-1}]
C_D,C_{De} = dimensionless wellbore storage
d = well distance
e_{D1},e_{D2},
e_{D3} = off-centering parameters
h = formation thickness, m [ft]
k = permeability, md
L = well half-length, m [ft]
L_{D1},L_{D2},
L_{D3} = dimensionless lengths
m = semilog straight-line slope, kPa/cycle [psi/cycle]
M = measure point in the well
p_D = dimensionless pressure
p_i = initial pressure in reservoir, kPa [psi]
p_{wf} = wellbore pressure
Δp = pressure-drop variation
q = well flow rate, m^3/d [B/D]
r_e = radius of drainage of well
r_w = well radius
r_w' = effective well radius
s,s',s_g = skin (global, real, geometrical)
t = time, hours
t_D = dimensionless time
x,y,z = coordinates
x_e,y_e = dimensions of reservoir
y_D = dimensionless position of point M
z_w = vertical coordinate of center of horizontal well
α = geometrical skin
δ = distance from point of measurement to end of well
μ = viscosity, mPa·s [cp]
ϕ = porosity

Subscripts

D = dimensionless
s = skin

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Appendix A—Horizontal Well With Uniform Flow in an Infinite Homogeneous Reservoir

Analytical Solution. Pressure-drop variation vs. time at Point M in the well defined by the value of y can be analytically expressed with Green's function.[5,6] The Green's function corresponding to the solution for a horizontal well can be expressed as a product of the Green's functions of Ref. 5:

$$s(M,t)=\mathrm{I}(x)_{x_w=0}\mathrm{II}(y)_{y_w=0}\mathrm{VII}(z)_{z_e=h}.$$

TABLE A-1—SOLUTIONS

Infinite reservoir

$$p_D(t_D)=\frac{\sqrt{\pi}}{4}\int_0^{t_D}\frac{e^{-x_{wD}^2/4\tau_D}}{\sqrt{\tau_D}}\left(\operatorname{erf}\frac{1+y_D}{2\sqrt{\tau_D}}+\operatorname{erf}\frac{1-y_D}{2\sqrt{\tau_D}}\right)\left[1+2\sum_{n=1}^{\infty}e^{-n^2\pi^2L_{D3}^2\tau_D}\cos^2(n\pi e_{D3})\right]d\tau_D. \quad \text{(A-1)}$$

Limited reservoir (impermeable boundary)

$$p_D(t_D)=2\pi L_{D1}L_{D2}\int_0^{t_D}\left\{1+2\sum_{n=1}^{\infty}e^{-n^2\pi^2L_{D1}^2\tau_D}\cos(n\pi e_{D1})\cos[n\pi(e_{D1}+x_{wD}L_{D1})]\right\}\left\{1+\frac{2}{\pi L_{D2}}\sum_{n=1}^{\infty}\right.$$

$$\left.\times\frac{1}{n}e^{-n^2\pi^2L_{D2}\tau_D}\sin(n\pi L_{D2})\cos(n\pi e_{D2})\cos[n\pi(e_{D2}+y_DL_{D2})]\right\}\left[1+2\sum_{n=1}^{\infty}e^{-n^2\pi^2L_{D3}\tau_D}\cos^2(n\pi e_{D3})\right]d\tau_D. \quad \text{(A-6)}$$

Limited reservoir (constant-pressure boundary)

$$p_D(t_D)=8L_{D1}\int_0^{t_D}\left\{\sum_{n=1}^{\infty}e^{-n^2\pi^2L_{D1}^2\tau_D}\sin(n\pi e_{D1})\sin[n\pi(e_{D1}+x_{wD}L_{D1})]\right\}\left\{\sum_{n=1}^{\infty}\frac{1}{n}e^{-n^2\pi^2L_{D2}^2\tau_D}\sin(n\pi L_{D2})\right.$$

$$\left.\times\sin(n\pi e_{D2})\sin[n\pi(e_{D2}+y_DL_{D2})]\right\}\left[1+2\sum_{n=1}^{\infty}e^{-n^2\pi^2L_{D3}^2\tau_D}\cos^2(n\pi e_{D3})\right]d\tau_D. \quad \text{(A-7)}$$

As a result, pressure drop can be written as follows:

$$p(M,t)=\frac{qB}{2L\phi c_t}\int_0^t s(M,\tau)d\tau.$$

This solution gives the pressure drop in an anisotropic reservoir. To obtain the pressure in the well, we will later assume the reservoir to be isotropic.

Pressure drop in the well with dimensionless parameters is given by Eq. A-1 of Table A-1. This equation is unchanged by replacing y_D by $-y_D$ or e_{D3} by $1-e_{D3}$. To simplify, we will call $f_1(\tau_D)$, $f_2(\tau_D)$, and $f_3(\tau_D)$ the three terms used in the integral of Eq. A-1.

Approximate Solutions—Time Criteria. ***Small Time Values.*** The asymptotic behavior of f_1, f_2, and f_3 when τ_D approaches zero are known. δ is the distance from the point of measurement $M(x_D)$ to the nearest end of the well. For small values of τ_D^2,

$$f_2(\tau_D)\cong 2 \text{ for } y_D\neq 1 \;[kt/(\phi\mu c_t\delta^2)<1/20],$$

$$f_2(\tau_D)\cong 1 \text{ for } y_D=1 \;[kt/(\phi\mu c_t\delta^2)<1/20],$$

and

$$f_3(\tau_D)\cong 1/(2L_{D3}\sqrt{\pi\tau_D}).$$

The approximate solution of Eq. A-1 for small values of τ_D and $y_D\neq 1$ (vertical radial flow) can be written as follows:

$$p_D(t_D)=\frac{1}{4L_{D3}}\int_0^{t_D}\frac{e^{-x_{wD}^2/4\tau_D}}{\tau_D}d\tau_D$$

$$=\frac{1}{2L_{D3}}\left[-\frac{1}{2}\mathrm{Ei}\left(-\frac{x_{wD}^2}{4t_D}\right)\right]. \quad \text{(A-2)}$$

When $y_D=1$, the coefficient $1/(2L_{D3})$ becomes $1/(4L_{D3})$.

The approximation of function $f_2(\tau_D)$ is no longer valid when end effect is perceptible. The criterion mentioned above relates to function $f_2(\tau_D)$. It appears to be too strict when applied to function p_D. From the analysis of the curves generated by Eq. A-1, it appeared that a better criterion for p_D could be obtained by substituting 1/6 for 1/20. Therefore, we should consider that the end effect is perceptible when $t_D>(1-y_D)^2/6$ with Darcy units.

The approximation of $f_3(\tau_D)$ is not valid when the nearest boundary affects well pressure. As long as the vertical radial flow exists, the problem becomes that of a vertical well at a distance d from a sealing fault. In this case, the formula of the radius of investigation, $kt/(\phi\mu c_t d^2)=0.25$, gives values too small for time. For this reason, we prefer $kt/(\phi\mu c_t d^2)=0.32$. The nearest boundary will have an effect on a horizontal well only for

$$t_D>0.32\left(\frac{1-e_{D3}}{L_{D3}}\right)^2 \;(e_{D3}>0.5).$$

Large Time Values. For a large τ_D value, we have

$$f_1(\tau_D)\cong\frac{1}{\sqrt{\tau_D}},$$

$$f_2(\tau_D)\cong\frac{2}{\sqrt{\tau_D}},$$

and

$$f_3(\tau_D)\cong 1.$$

The solution of Eq. A-1 leads to the pseudoradial flow equation:

$$p_D(t_D)=0.5(\ln t_D+\alpha). \quad \text{(A-3)}$$

Geometrical Skin. Eq. A-3 refers to various phenomena that have to be distinguished (see Pseudoradial Flow). We assume that the real skin of the well, s, is zero.

With a pseudoradial flow, as defined in Flow Types Around a Horizontal Well, pressure drop is constant in the area where the well distorts the lines of flow, resulting in a negative pseudoskin (geometrical skin, s_g). The pseudoradial flow equation for $s=0$ can be written as follows:

$$p_i-p_{wf}(t)=\frac{qB\mu}{4\pi kh}\left(\ln\frac{kt}{\phi\mu c_t r_w^2}+0.809+2s_g\right). \quad \text{(A-4)}$$

During the pseudoradial flow, the behavior of a horizontal well with a geometrical skin, s_g, is similar to that of a vertical well whose radius is $r_w'=r_we^{-s_g}$. As we know, the relation between r_w' and

r_w in the case of a horizontal well with infinite conductivity in the center of the reservoir can be written as

$$r_w' = \frac{L(2\pi r_w/h)^{h/2L}}{1+\sqrt{1-(L/r_e)^2}},$$

where r_e is the radius of drainage of the well. Because flow is pseudoradial, we can suppose $L/r_e << 1$ and the above relation can be simplified.

So, the geometrical skin for a centered well can be inferred to be

$$s_g = \ln\frac{r_w}{r_w'} = \ln\frac{2x_{wD}}{(2\pi x_{wD}L_{D3})^{1/(2L_{D3})}}. \quad \text{(A-5)}$$

This equation applies to the case we are considering—i.e., to an infinite-conductivity well. It refers to a centered well. For a reasonable off-centering, however, the error range is low. Let us consider the case shown in Figs. 6 and 7, where off-centering is $e_{D3}=0.75$. We determined the geometrical skin from Eq. A-6, in which $L_{D3}=0.5$. Then we computed pressure values at the time of a pseudoradial flow with the logarithmic law (Eq. A-5). Comparing the computed values with those of Fig. 7 shows that the application of Eq. A-6 results in a pressure error on the order of 3% only. Practically, Eq. A-6 allows the determination of s_g with enough accuracy except in the case of wells whose off-centering is more than 0.75. In that case, the value obtained for s_g will have to be regarded as an order of magnitude.

Appendix B—Horizontal Well With Uniform Flow in a Limited Homogeneous Reservoir (Impermeable Boundaries)

Analytical Solution. An approach similar to the one described in Appendix A leads to the solution shown as Eq. A-6 in Table A-1 with the set of the additional parameters of Horizontal Well in a Homogeneous Reservoir With Impermeable Boundaries. The Green's function corresponding to the solution can be expressed as a product of the Green's functions of Ref. 5 as follows:

$$s(m,t) = \mathrm{VII}(x)_{x_w=0}\mathrm{X}(y)_{y_w=0}\mathrm{VII}(z)_{z_e=h}.$$

Remarks. By reason of symmetry, we can limit our approach to the following cases: $e_{D1}>0.5$, $e_{D2}>0.5$, and $e_{D3}>0.5$. Results are identical when e_{D1} is changed to $1-e_{D1}$ with x_{wD} replaced by $-x_{wD}$ and when e_{D2} is changed to $1-e_{D2}$ with y_D replaced by $-y_D$.

Note that when $e_{D2}\simeq 0.5$, pressure is the same at the point defined by y_D and $-y_D$. Therefore, a good simulation of an infinite-conductivity well can be obtained when the uniform-flow solution is used with $y_D=0.7$. This conclusion, however, cannot be extended to the cases in which $e_{D2}\neq 0.5$. For these cases, we can say that when y_D varies, the curves obtained rarely differ.

Appendix C—Horizontal Well With Uniform Flow in a Homogeneous Reservoir With Constant-Pressure Boundaries

The method used to solve this problem is quite similar to the one described in Appendix B. The solution is expressed by Eq. A-7 shown in Table A-1.

The Green's function corresponding to the solution can be expressed as a product of the Green's functions of Ref. 5 as follows:

$$s(M,t) = \mathrm{VIII}(x)_{x_w=0}\mathrm{XI}(y)_{y_w=0}\mathrm{VII}(z)_{z_e=h}.$$

The remarks of Appendix B apply to this case.

SI Metric Conversion Factors

atm	×	1.013 250*	E+05	=	Pa
bar	×	1.0*	E+05	=	Pa
bbl	×	1.589 873	E−01	=	m^3
cp	×	1.0*	E+00	=	mPa·s
ft	×	3.048*	E−01	=	m
in.	×	2.54*	E+00	=	cm
psi	×	6.894 757	E+00	=	kPa
psi^{-1}	×	1.450 377	E−04	=	Pa^{-1}

*Conversion factor is exact.

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